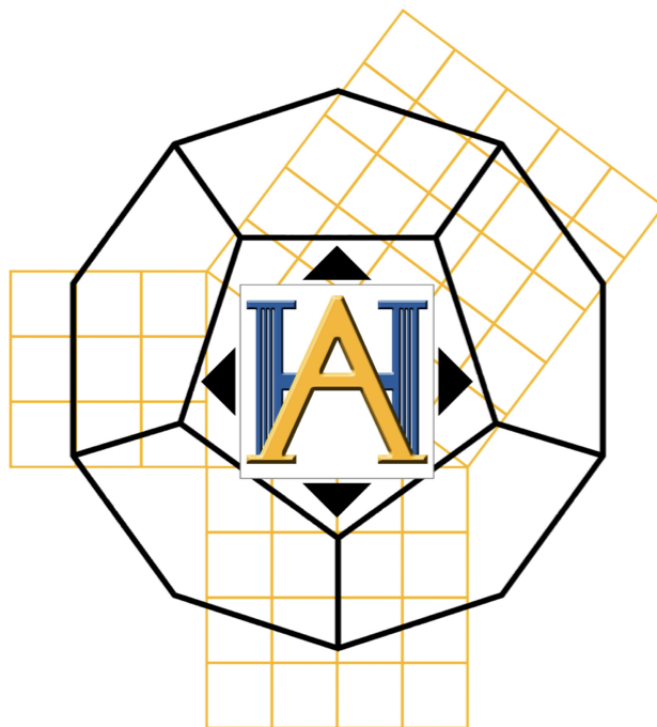




**AMERICAN HERITAGE SCHOOLS
COMPETITIVE MATHEMATICS**

**2024 Mock Theta
Probability & Combinatorics**

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Instructions: Unless specified, assume that all draws are completely random and are done without replacement. Questions typically increase in difficulty the further they are on the test, but may vary occasionally. NOTA means “None of the Above”.

~~~~~ Good luck and have fun! ~~~~~

**Use the following information for the entire test:**

Unlike a standard deck of 52 cards, a standard Tarot deck consists of 78 cards. 22 of these cards are the *Major Arcana* and they are labeled starting from 0 and ending at *XXI*. The other 56 cards are the *Minor Arcana* and similarly to standard decks, consist of 4 suits: *Wands*, *Swords*, *Cups*, and *Pentacles*. Unlike a standard deck, however, each suit consists of 14 ranks: 10 pip, or number, cards and 4 court, or face, cards. The *Major Arcana* are listed during the test with their numbering being found next to them in roman numerals.

1. The *Fool* (0) is the first of the *Major Arcana* and the other *Major Arcana* represents the *Fool's* journey. In a standard Tarot deck, what is the probability that the 78<sup>th</sup> card you draw is the *Fool*?

(A)  $\frac{1}{52}$  (B)  $\frac{1}{52}$  (C)  $\frac{1}{78}$  (D) 0 (E) NOTA

2. The *Magician* (I) follows the *Fool* as the second *Major Arcana*. How many unique permutations of *MAGICIAN* are there?

(A) 6720 (B) 10080 (C) 20160 (D) 40320 (E) NOTA

3. Next is the *High Priestess* (II), which represents one's spiritual instincts. Similarly, the *Cups* suit, representative of clergymen, also relates to spiritual matters. In a standard Tarot deck, what is the probability of drawing either a *Cups*, a court card, or the *High Priestess*?

(A)  $\frac{3}{78}$  (B)  $\frac{8}{39}$  (C)  $\frac{27}{78}$  (D)  $\frac{31}{78}$  (E) NOTA

4. The fourth and fifth *Major Arcana* are the *Empress* (III) and the *Emperor* (IV) respectively. A Queen and a King plan to meet in a café from 2 P.M. to 4 P.M. but they each only stay for 30 minutes at a time before leaving. If both of them arrive at a random time from 2 P.M. to 4 P.M., what is the probability that the two of them meet each other in the café?

(A)  $\frac{5}{16}$  (B)  $\frac{7}{16}$  (C)  $\frac{9}{16}$  (D)  $\frac{11}{16}$  (E) NOTA

5. The *Hierophant* (V) is next. Now suppose we are playing *Poker* with a deck of *Minor Arcana*. A *Flush* is when one draws 5 cards of the same suit. If we draw 5 cards, what is the probability that we draw a *Flush*?

(A)  $4 \left( \frac{13! 47!}{8! 52!} \right)$  (B)  $\frac{14! 73!}{9! 78!}$  (C)  $\frac{14! 51!}{9! 56!}$  (D)  $4 \left( \frac{14! 51!}{9! 56!} \right)$  (E) NOTA

6. The *Lovers* (VI) is the seventh *Major Arcana* and, unsurprisingly, represents relationships and romances. 15 couples attend a party and each person gives a handshake to every other person who is not their partner, whom they give a kiss to. Let  $H$  be the number of handshakes that occur and let  $K$  be the number of kisses that occur. Find  $H - K$ .

(A) 405 (B) 415 (C) 420 (D) 435 (E) NOTA

7. Following the *Lovers* is the *Chariot* (VII). In *Dungeons and Dragons*, a dice roll for a standard, twenty-sided die (D20) dictates the amount of damage one deals in an attack, where the amount of damage given is equal to the number rolled. Suppose we can't deal more damage in a single attack than the enemy's health points (HP). For example, if we attack an enemy with 5 HP and rolled a 7, we would still deal 5 damage. If we attack an enemy with 16 HP, what is the expected amount of damage we would deal?
- (A) 10                      (B) 10.25                      (C) 10.5                      (D) 10.75                      (E) NOTA
8. The ninth *Major Arcana* is the *Strength* (VIII). What is the number of 4 digit positive integers less than 5200 such that the ones digit is a perfect square and the tens digit is a prime?
- (A) 504                      (B) 600                      (C) 636                      (D) 672                      (E) NOTA
9. After that is the *Hermit* (IX). What are the last 3 digits of  $11^{99}$ ? *Hint: It may be useful to consider binomial expansion.*
- (A) 001                      (B) 091                      (C) 901                      (D) 991                      (E) NOTA
10. The *Wheel of Fortune* (X) is the eleventh *Major Arcana* and represents one's luck. Suppose you are in a gameshow, where you are presented with 3 doors and you must pick one of them to go through. Behind 1 door is the grand prize while behind each of the other 2 is nothing. After picking a door, the host opens one of the unopened doors, always revealing that there is nothing behind it and presents you with the option to switch to a different door. If you play optimally, what is the probability that you will end up at the door with the grand prize?
- (A)  $\frac{1}{3}$                       (B)  $\frac{1}{2}$                       (C)  $\frac{7}{12}$                       (D)  $\frac{2}{3}$                       (E) NOTA
11. Then is the *Justice* (XI). Suppose I have a magical 6-sided die such that whenever I roll a value, I cannot roll it again. What is the expected value of the 6<sup>th</sup> dice roll?
- (A) 2                      (B) 2.5                      (C) 3                      (D) 3.5                      (E) NOTA
12. The next *Major Arcana* is the *Hanged Man* (XII). Suppose I have a  $10 \times 5$  grid of unit squares. How many rectangles are in this grid?
- (A) 50                      (B) 576                      (C) 825                      (D) 1386                      (E) NOTA
13. The number "13" is often associated with some kind of misfortune and many have a superstition about it. Tarot cards back this superstition up, as the *Death* (XIII) is the numbered 13 in most Tarot decks. What proportion of the factors of  $13!$  are divisible by 3?
- (A)  $\frac{1}{3}$                       (B)  $\frac{5}{6}$                       (C)  $\frac{4}{5}$                       (D)  $\frac{7}{8}$                       (E) NOTA
14. Then there is the *Temperance* (XIV). At the Frazer School of Comedic Arrangements, its 72 students take at least one of 3 classes: Stand-Up Comedy, Theatrics, and Mime Class. 10 people only take Stand-Up Comedy and 14 people only take Theatrics. 11 people take both Theatrics and Stand-Up Comedy, 23 people take both Theatrics and Mime Class, and 5 people take all three classes. How many people take Mime Class but don't take Theatrics?
- (A) 15                      (B) 17                      (C) 19                      (D) 21                      (E) NOTA

15. The next card, the *Devil* (XV), represents addictions. In *Les Gevas*, people play on slot machines that have 6 different jackpots, each with the same probability of occurring. Like the avid gambler he is, Gée tries to get every jackpot on these machines. What is the expected number of jackpots Gée needs to get in order to get all the different jackpots?
- (A) 14.5 (B) 14.7 (C) 14.9 (D) 15 (E) NOTA
16. Following that, the next card is the *Tower* (XVI). Gée throws a dart at dartboard of radius 1 such that where ever it lands on this dartboard is random and records the distance from the dart to the center of the dartboard. Neng selects a random number in the interval  $[0, 1]$ . What is the probability Gée's number less than Neng's number? *Hint: Consider trying to describe this situation with a shape that can represent both a circle and a line.*
- (A)  $\frac{2}{3}$  (B)  $\frac{1}{6}$  (C)  $\frac{1}{2}$  (D)  $\frac{1}{3}$  (E) NOTA
17. The *Star* (XVII) is obviously going to have a Stars-and-Bars question. Suppose  $x, y$ , and  $z$  are non-negative integers. Find the number of ordered triples  $(x, y, z)$  that satisfy  $4x + y + 4z = 42$ .
- (A) 32 (B) 54 (C) 66 (D) 84 (E) NOTA
18. The *Moon* (XVIII) and the *Sun* (XIX) are the next cards. Gée and Kary are in a best of 5 Rock-Paper-Scissors competition, where due to his superior skill, Gée has a 0.6 chance of winning each match. Given that Gée won, what is the probability that he did so flawlessly? A flawless win means that Gée did not lose a single match in this best of 5 set.
- (A)  $\frac{5}{13}$  (B)  $\frac{25}{39}$  (C)  $\frac{13}{25}$  (D)  $\frac{27}{125}$  (E) NOTA
19. The *Judgement* (XX) is the second to last of the *Major Arcana*. Let  $\triangle ABC$  have  $m\angle A = 30^\circ$ . Sides  $AB$  and  $AC$  are of random length.  $AB$  has a side length of  $n$  with a probability of  $\frac{1}{2^n}$  while  $AC$  has a side length of  $2k$  with a probability of  $\frac{2}{3^k}$  for positive integers  $n$  and  $k$ . What is the expected value of the area of  $\triangle ABC$ ?
- (A) 1.5 (B) 2 (C) 3 (D) 6 (E) NOTA
20. Congratulations on making it to the end of the test! *The World* (XXI) is final *Major Arcana* and represents the end of the *Fool's* journey. What is the probability of drawing 22 distinct *Major Arcana* in a row, followed by the *The World*?
- (A) 0 (B)  $\frac{22! \cdot 56!}{78!}$  (C)  $\frac{23! \cdot 55!}{78!}$  (D)  $\left(\frac{23}{78}\right)^{23}$  (E) NOTA